Project Planning & Management Architecture Report

ENTERPRISE DATA ENGINEERING & TELECOM CHURN RETENTION FRAMEWORK

> **Executive Planning Charter & Scope**
>
> **Track & Domain:** Data Engineering Track (AI & Data Science) – Round 5
> **Dataset Dimensions:** 100,000 instances × ~100 structural features (Demographics + Interaction Records)
> **Core Architecture:** SQL ELT Ingestion → ML Classification Pipelines → SHAP XAI → Streamlit App Deployment

1. PROJECT CHARTER & HIGH-LEVEL OBJECTIVES

Customer attrition (churn) is a pivotal revenue threat in the telecommunications sector. Acquiring a new subscriber is 5 to 7 times more expensive than retaining an existing one. In competitive markets such as Egypt's MNO landscape (with over 127 million mobile subscriptions and active Mobile Number Portability), early identification of high-risk customers protects Average Revenue Per User (ARPU).

This Project Planning & Management Report outlines the governance, resource allocation, technical workflows, and timeline necessary to execute an enterprise-grade churn prediction system. The technical blueprint bridges relational database engineering (SQL ELT), imbalanced machine learning modeling (XGBoost/RF), post-hoc explainability (SHAP), and real-time interactive web deployment (Streamlit/Flask).

2. WORK BREAKDOWN STRUCTURE (WBS) & MILESTONE ALIGNMENT

The project execution strictly follows the official competition roadmap divided into three core Milestones:

- **Milestone 1: Data Collection, Exploration & Preprocessing:** Local SQL database setup, relational joins across `Client.csv` and `Record.csv` (100k rows), categorical encoding, missing value imputation, and class imbalance reports.
- **Milestone 2: Predictive Model Development & Interpretation:** Benchmarking Logistic Regression, Random Forest, and XGBoost; SMOTE oversampling; hyperparameter tuning; and SHAP feature importance analysis.
- **Milestone 3: Deployment & Retention Strategy:** Building an interactive Streamlit/Flask web interface, mapping risk probabilities to targeted retention offers, and producing executive PDF/presentation deliverables.

3. RESPONSIBILITY ASSIGNMENT MATRIX (RACI MATRIX)

To ensure clear ownership and seamless execution, responsibilities across data engineering, machine learning development, and business strategy are mapped below:

Project Deliverable / Task	Data Engineer	ML Engineer	Business Analyst
SQL Database Ingestion & ELT Joins	Accountable (A) / Responsible (R)	Consulted (C)	Informed (I)
EDA, Viz & Class Imbalance Report	Responsible (R)	Accountable (A)	Consulted (C)
Feature Preprocessing & Scaling	Responsible (R)	Accountable (A)	Informed (I)
Model Training (Logistic, RF, XGBoost)	Informed (I)	Accountable (A) / Responsible (R)	Consulted (C)
Hyperparameter Tuning & SHAP Analysis	Informed (I)	Accountable (A) / Responsible (R)	Consulted (C)
Streamlit / Flask Web Deployment	Consulted (C)	Accountable (A) / Responsible (R)	Consulted (C)
Data-Backed Retention Strategy	Informed (I)	Consulted (C)	Accountable (A) / Responsible (R)
Final Presentation & Report Submission	Responsible (R)	Responsible (R)	Accountable (A)

4. EXECUTION SCHEDULE & DETAILED GANTT TIMELINE

The project is structured strictly over a 3 to 4-week execution timeline, adaptable for either a solo practitioner or a collaborative team of 5 members:

- **Week 1 (Milestone 1):** Relational schema creation in SQLite/PostgreSQL, chunked SQL ingestion of 100k rows, exploratory data visualizations (bar plots, histograms), and feature scaling.
- **Week 2 (Milestone 2):** Split train/test datasets, execute SMOTE oversampling, train classification algorithms, prioritize Recall/ROC-AUC evaluation metrics, and extract SHAP feature importance plots.
- **Weeks 3–4 (Milestone 3):** Build Streamlit / Flask UI interface for real-time risk scoring, formulate data-backed retention recommendations (discount tiers, `eqpdays` handset upgrades), and assemble final executive presentation slides.

5. RISK MANAGEMENT & TECHNICAL GOVERNANCE FRAMEWORK

Proactive risk management ensures system stability, memory efficiency, and commercial alignment during production execution:

Risk Category	Identified Risk Event	Proactive Mitigation Strategy
Memory Overflow	RAM exhaustion when processing 100,000 rows × 100 features in Colab.	Use chunked SQL reads (`chunksize=10000`), downcast data types, and leverage Parquet format.
Model Overfitting	Tree-based models (XGBoost) memorizing noise on minority churn class.	Enforce Stratified K-Fold Cross-Validation, apply hyperparameter regularization (`max_depth`, `subsample`).
Imbalance Bias	High accuracy but near-zero Recall on minority churners (~15% baseline).	Prioritize Recall and ROC-AUC metrics; apply SMOTE oversampling and cost-sensitive class weights.
Deployment Lag	Streamlit app latency during real-time inference rendering.	Pre-serialize trained pipelines via `joblib`; optimize session state variables in Streamlit.
Business Misalignment	Technical predictions failing to translate into practical retention campaigns.	Map high-risk churn drivers directly to actionable business offers (e.g., equipment upgrades for high `eqpdays`).

6. PRESCRIPTIVE RETENTION STRATEGY MAPPING

To bridge technical model outputs with operational business decisions, predicted risk scores trigger automated, data-backed customer interventions:

- **High-Value / High-Risk Segment:** Dispatch subsidized equipment upgrade packages targeting subscribers with long device tenure (`eqpdays > 365`), backed by personalized data bonuses.
- **Price-Sensitive / Low Usage Segment:** Offer seamless migration to flexible hybrid billing tiers to counteract macroeconomic pressures and competitive mobile number portability (MNP) offers.

7. DELIVERABLES CHECKLIST & QUALITY ASSURANCE

The final submission package adheres strictly to the official contest guidelines and quality benchmarks across all three milestones:

Milestone	Required Official Deliverables	Quality Assurance Criteria
Milestone 1	• Cleaned, combined dataset ready for ML • EDA Jupyter Notebook with charts & histograms • Customer behavior summary report	No missing values; explicit handling of class imbalance; verified SQL join key consistency.
Milestone 2	• Trained classification model binaries (`.joblib`) • Model evaluation metrics report (ROC-AUC/Recall) • Feature importance & SHAP plots	Cross-validated metrics; clear prioritization of minority Recall over raw accuracy.
Milestone 3	• Live Web Application (Deployed Streamlit / Flask Interface) • Retention Strategy Document (Data-backed mitigation roadmap) • Executive Presentation (Final slides PDF)	Interactive real-time prediction UI; clear ROI mapping for retention offers; valid source links.

SUMMARY & PROJECT SIGN-OFF

This Project Planning & Management framework provides an end-to-end operational blueprint. By combining SQL ELT data pipelines, robust machine learning modeling, SHAP explainability, and real-time Streamlit deployment, the project guarantees both technical excellence and actionable business impact for telecommunications subscriber retention.